

Ch. 0 - Presentation

Secure software development and web security

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Haute École Bruxelles-Brabant
École supérieure d'Informatique



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Introduction

The teacher

■ Romain Absil

- `rabsil+student@he2b.be`
- `romain.absil+student@ulb.be`
- There is no mistake with '+'

■ If you have a question, ask

- This is a bite-free class

■ Not so formal sessions

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The course

Pace

- Dispensed 4h per week
- Labs and theory together
 - Sometimes lab, sometimes theory
- The first two chapters are assumed to be recalls
- “Optional” presence
 - ... but take responsibility
- You’re encouraged to come with a computer if you have one

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Objectives

- Understand several attacks on system and their applications
 - Buffer overflow, data remanence, injection, etc.
- *Design* software to be secure
 - Building protocols using cryptographic hash functions, cipher algorithms, etc.
 - Math left aside here
- Use security libraries
 - Honestly, choose what you want
- Implementation of several concepts

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References and git repositories

- W. Du: Computer Security - A hands-on approach, CreateSpace Independent Publishing Platform; 1 edition (2017)
- K. G. Mehrotra, C. K. Mohan and H. Huang: Anomaly detection - Principles and algorithms, Springer (2017)
- C. Paar and J. Pelzl: Understanding Cryptography - A Textbook for Students and Practitioners, Springer (2010)
- B. Schneier : Secrets and Lies: Digital Security in a Networked World - Wiley, 15th Anniversary edition (2015)
- OWASP: <https://owasp.org/>, last accessed on June 22, 2024
- NIST: <https://www.nist.gov/>, last accessed on June 22, 2024
- Public repositories
 - Public group: <https://gitlab.com/computer-security3>
 - `sec1` and `sec2` are assumed to be covered during your bachelor's degree
 - `sec3` is the course, with `labs` and `project` for the practical part

Grading you

January

- **35% continuous evaluation**

- 2 or 3 labs

- **65% exam**

- Project defense on exam day

- **Labs**

- Rainbow attack

- Password attack on naive password storage with hash functions

- Buffer overflow

- A set of "mini-me" challenges

- Race conditions and shells (maybe)

- A set of remote attacks by passing secure context

- **Computer project**

- Web application enforcing end-to-end encryption

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- A set of remote attacks by passing secure context

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 - Project defense on exam day
- Labs
 - Rainbow attack
 - Password attack on naive password storage with hash functions
 - Buffer overflow
 - A set of "real-world" challenges
 - Race conditions and shells (maybe)
 - Shellshock: A set of remote attacks by passing secure context
- Computer project
 - Web application enforcing end-to-end encryption

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 - Project defense on exam day
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 - Rainbow attack
 - Password attack on naive password storage with hash functions
 - Buffer overflow
 - A set of "mini-cms" challenges
 - Race conditions and shells (maybe)
 - ...etc. A set of various attacks to passing secure content
- Computer project
 - Web application enforcing end-to-end encryption

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 - Attack on naive password storage with hash functions
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August

■ Several options

- 1 Usually another project
 - 2 Fix (alone) what was wrong in January
 - Labs and project
 - 3 Make a report on a scientific topic not covered by the course
 - 4 Oral exam (theoretic questions)
- I make the choice, depending on the number of students who failed

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